Week 9 Homework

This is a preview of the published version of the quiz

Started: Jul 2 at 7:50am

Quiz Instructions

Question 1

1 pts

(Lesson 7.1: Introduction to Random Variate Generation.) Unif(0,1) PRNs can be used to generate which of the following random entities?

- a. Exp(λ) random variates
- b. Nor(0,1) random variates
- c. Triangular random variates
- d. Bern(p) random variates
- e. Nonhomogeneous Poisson processes
- f. All of the above --- and just about anything else!

Question 2

1 pts

(Lesson 7.2: Inverse Transform Theorem --- Intro.) If X is an Exp(λ) random variable with c.d.f. $F(x) = 1 - e^{-\lambda x}$, what's the distribution of the random variable $1 - e^{-\lambda X}$?

- a. Unif(0,1)
- b. Nor(0,1)
- c. Triangular
- d. Exp(λ)

- e. None of the above

Question 3

1 pts

(Lesson 7.2: Inverse Transform Theorem --- Intro.) If U is a Unif(0,1) random variable, what's the distribution of $-\frac{1}{\lambda}\ln(U)$?

- a. Unif(0,1)
- b. Nor(0,1)
- c. Triangular
- d. Exp(λ)
- e. None of the above

Question 4

1 pts

(Lesson 7.2: Inverse Transform Theorem --- Intro.) Suppose that $U_1, U_2, \ldots, U_{5000}$ are i.i.d. Unif(0,1) random variables. Using Excel (or your favorite programming language), simulate $X_i = -\ln(U_i)$ for $i = 1, 2, \ldots, 5000$. Draw a histogram of the 5000 numbers. What p.d.f. does the histogram look like?

- a. Uniform
- b. Normal
- c. Triangular
- d. Exponential
- e. Bernoulli

Question 5

1 pts

(Lesson 7.3: Inverse Transform --- Continuous Examples.) Suppose the c.d.f. of X is $F(x) = x^3/8$, $0 \leq x \leq 2$. Develop a generator for X and demonstrate with $U = 0.54$.

- a. $X = U^3/8 = 0.0197$
- b. $X = 8U^3 = 1.260$
- c. $X = 2U^{1/3} = 1.629$
- d. $X = 4U^{1/3} = 3.257$
- e. $X = 8U^{1/3} = 6.515$

Question 6

1 pts

(Lesson 7.3: Inverse Transform --- Continuous Examples.) If X is a Nor(0,1) random variate, and $\Phi(x)$ is the Nor(0,1) c.d.f., what is the distribution of $\Phi(X)$?

- a. Uniform
- b. Normal
- c. Triangular
- d. Exponential
- e. Bernoulli

Question 7

1 pts

(Lesson 7.3: Inverse Transform --- Continuous Examples.) If U is a Unif(0,1) random variate, and $\Phi(x)$ is the Nor(0,1) c.d.f., what is the distribution of $2\Phi^{-1}(U) + 3$?

- a. Unif(0,1)
- b. Unif(3,2)

- c. Nor(0,1)
- d. Nor(3,2)
- e. Nor(3,4)

Question 8

1 pts

(Lesson 7.4: Inverse Transform --- Discrete Examples.) How would you simulate the sum of two 6-sided dice tosses? (Note that $\lceil \cdot \rceil$ is the round-up function; and all of the U's denote PRNs.)

- a. $12U$
- b. $\lceil 12U \rceil$
- c. $6U_1 + 6U_2$
- d. $\lceil 6U_1 \rceil + \lceil 6U_2 \rceil$
- e. None of the above

Question 9

1 pts

(Lesson 7.4: Inverse Transform --- Discrete Examples.) If U is Unif(0,1), how can we simulate a Geom(0.6) random variate?

- a. $\lceil \ln(U)/\ln(0.4) \rceil$
- b. $\lceil \ln(1-U)/\ln(0.4) \rceil$
- c. $\lceil \ln(U)/\ln(0.6) \rceil$
- d. $\lceil \ln(1-U)/\ln(0.6) \rceil$
- e. Both (a) and (b)
- f. Both (c) and (d)

Question 10

1 pts

(Lesson 7.6: Convolution.) Suppose that U and V are PRNs. Let $X = U + V$. Simulate this 5000 times, and draw a histogram of the 5000 numbers. What p.d.f. does the histogram look like?

- a. Uniform
- b. Normal
- c. Triangular
- d. Exponential
- e. Bernoulli

Question 11

1 pts

(Lesson 7.6: Convolution.) Suppose that $U_1, U_2, \ldots, U_{24}$ are i.i.d. PRNs. What is the approximate distribution of $X = 5 + 3\sum_{i=1}^{24} U_i$?

- a. Uniform
- b. Nor(0,1)
- c. Nor(5,1)
- d. Nor(12,2)
- e. Nor(41,6)
- f. Nor(41,18)

Question 12

1 pts

(Lesson 7.6: Convolution.) If U_1, U_2, U_3 are PRNs, what's the distribution of $-2\ln(U_1^2(1-U_2)^2U_3^2)$?

- a. Exp(1/2)
- b. Exp(4)
- c. Erlang$_2(1/2)$
- d. Erlang$_3(1/4)$
- e. Erlang$_3(2)$

Question 13

1 pts

(Lesson 7.7: Acceptance-Rejection --- Intro.) In general, the majorizing function $t(x)$ is itself a p.d.f. $f(x)$.

- True
- False

Question 14

1 pts

(Lesson 7.9: Acceptance-Rejection --- Continuous Examples.) Suppose that X is a continuous RV with p.d.f. $f(x) = 30x^4(1-x)$, for $0 < x < 1$. What's a good method that you can use to generate a realization of X?

- a. Inversion
- b. Convolution
- c. Box-Muller
- d. Acceptance-Rejection
- e. Composition

Question 15

1 pts

(Lesson 7.8: Acceptance-Rejection --- Proof.) Consider the constant $c = \int_{\mathbb{R}} t(x)\, dx = 5$. On average, how many iterations (trials) will the A-R algorithm require?

- a. 1/5
- b. 5
- c. 10
- d. 25
- e. None of the above

Question 16

1 pts

(Lesson 7.10: Acceptance-Rejection --- Poisson Distribution.) Suppose that $U_1 = 0.65$, $U_2 = 0.45, U_3 = 0.82, U_4 = 0.09$, and $U_5 = 0.26$. Use our acceptance-rejection technique from class to generate $N \sim \text{Pois}(\lambda = 3.7)$. (You may not need to use all of the uniforms.)

- a. N=0
- b. N=1
- c. N=2
- d. N=3
- e. N=4

Not saved

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